



DEPARTMENT OF PHYSICS

TFY4230 - STATISTICAL PHYSICS

Numerical project: Gas simulation

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1 Dimensionless equations

Equations of motion

We consider N particles with positions $\mathbf{r}_i = (x_i, y_i)$. The motion of the particles is governed by Newton's second law.

$$m \frac{d^2 \mathbf{r}_i}{dt^2} = -\frac{\partial V}{\partial \mathbf{r}_i}. \quad (1)$$

The total potential energy consists of particle-particle interactions and particle-wall interactions.

$$V = \sum_{i < j} V_{LJ}(r_{ij}) + \sum_i V_{wall}(r_i) \quad (2)$$

Wall interaction

The wall force is obtained from the wall potential,

$$F_{x,i} = -\frac{\partial V_{wall}}{\partial x_i},$$

and is only present when $r_i > R$. This result in the simplification of $F_{x,i}$, yielding:

$$F_{x,i} = -\frac{\partial V_{wall}}{\partial x_i} = -K(r_i - R) \frac{\partial r_i}{\partial x_i}.$$

By expressing r_i in cartesian coordinates it is possible to solve the expression further, with

$$r_i = \sqrt{x_i^2 + y_i^2},$$

resulting in

$$\frac{\partial r_i}{\partial x_i} = \frac{x_i}{r_i}.$$

The simplified equation for the wall force is expressed as

$$F_{x,i} = -K(r_i - R) \frac{x_i}{r_i}. \quad (3)$$

Particle Interaction

The Lennard-Jones potential is

$$V_{LJ}(r_{ij}) = 4\epsilon \left[\left(\frac{a}{r_{ij}} \right)^{12} - \left(\frac{a}{r_{ij}} \right)^6 \right]. \quad (4)$$

The force between two particles is obtained from the potential as

$$F(r_{ij}) = -\frac{dV_{LJ}}{dr_{ij}}. \quad (5)$$

Differentiating with respect to r_{ij} gives

$$\frac{dV_{LJ}}{dr_{ij}} = 4\varepsilon \left[-12 \frac{a^{12}}{r_{ij}^{13}} + 6 \frac{a^6}{r_{ij}^7} \right].$$

The resulting force is

$$F(r_{ij}) = 24\varepsilon \left[2 \left(\frac{a}{r_{ij}} \right)^{12} - \left(\frac{a}{r_{ij}} \right)^6 \right] \frac{1}{r_{ij}}. \quad (6)$$

The x -component becomes

$$F_{x,i} = 24\varepsilon \left[2 \left(\frac{a}{r_{ij}} \right)^{12} - \left(\frac{a}{r_{ij}} \right)^6 \right] \frac{x_i - x_j}{r_{ij}^2}. \quad (7)$$

Combining both interactions gives Newton's equation as described in formula (9) in the assignment.

$$m \frac{d^2 x_i}{dt^2} = -K(r_i - R) \frac{x_i}{r_i} + \sum_{j \neq i} 24\varepsilon \left[2 \left(\frac{a}{r_{ij}} \right)^{12} - \left(\frac{a}{r_{ij}} \right)^6 \right] \frac{x_i - x_j}{r_{ij}^2}. \quad (8)$$

1.1 Dimensionless equations

We introduce the characteristic time scale

$$\tau = \sqrt{\frac{ma^2}{\varepsilon}}. \quad (9)$$

Dimensionless variables are defined as

$$x_i = a\tilde{x}_i, \quad y_i = a\tilde{y}_i, \quad t = \tau\tilde{t}.$$

The distance scale as

$$r_i = a\tilde{r}_i, \quad r_{ij} = a\tilde{r}_{ij}.$$

Using the chain rule we get

$$\begin{aligned} \frac{d}{dt} &= \frac{1}{\tau} \frac{d}{d\tilde{t}}, \\ \frac{d^2}{dt^2} &= \frac{1}{\tau^2} \frac{d^2}{d\tilde{t}^2}. \end{aligned}$$

Substituting the dimensionless variables $x_i = a\tilde{x}_i$, $r_i = a\tilde{r}_i$, and $t = \tau\tilde{t}$ into Newton's equation and using the relations above gives the dimensionless equation

$$\frac{d^2 \tilde{x}_i}{d\tilde{t}^2} = -\tilde{K}(\tilde{r}_i - \tilde{R}) \frac{\tilde{x}_i}{\tilde{r}_i} + \sum_{j \neq i} 24 \left[2 \left(\frac{1}{\tilde{r}_{ij}} \right)^{12} - \left(\frac{1}{\tilde{r}_{ij}} \right)^6 \right] \frac{\tilde{x}_i - \tilde{x}_j}{\tilde{r}_{ij}^2}. \quad (10)$$

where,

$$\tilde{R} = \frac{R}{a}, \quad \tilde{K} = \frac{Ka^2}{\varepsilon}. \quad (11)$$

The dimensionless formulation removes the physical parameters m , a , and ε . The dynamics is therefore controlled only by the wall stiffness $\tilde{K}$ and the container size $\tilde{R}$.

2 Numerical solver

To solve equation 10 we will implement the velocity Verlet algorithm, given by

$$\mathbf{r}_i(t + \Delta t) = \mathbf{r}_i(t) + \mathbf{v}_i(t) \Delta t + \frac{1}{2} \frac{\mathbf{f}_i(t)}{m} \Delta t^2 \quad (12)$$

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) + \frac{\mathbf{f}_i(t) + \mathbf{f}_i(t + \Delta t)}{2m} \Delta t. \quad (13)$$

The implementation can be found in the attached document, and for the given parameters $N = 10$, $\tilde{R} = 10$, $\tilde{K} = 25$, and $d\tilde{t} = 0,005$ the resulting dimensionless total energy is plotted as a function of time in figure 1. Note that to avoid discrepancies in the initial position we have distributed the particles along two circular layers, one at a radius of $\frac{\tilde{R}}{3}$ and another at $\frac{2\tilde{R}}{3}$.

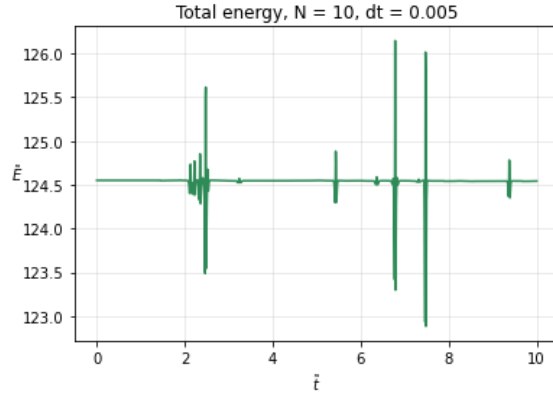


Figure 1: Dimensionless energy vs. time for the example parameters $N = 10$, $\tilde{R} = 10$, $\tilde{K} = 25$, and $d\tilde{t} = 0,005$.

We observe that the energy is fairly constant, with the exception of a few spikes here and there. This is likely due to the fact that the Lennard-Jones potential is quite stiff, due to the $\frac{1}{r^{12}}$ term. This, in combination with the time-discretizations in the Verlet algorithm, results in inconsistencies when particles come close together. We see that the energy settles back again after the particles gain distance between one another. All in all the solver seems to be working as expected.

The result of an analysis of the relative energy difference can be found in figure 2, where the parameters are equivalent to those used earlier, but we vary dt . We observe that the energy spikes become significantly smaller for finer time-steps, just as expected.

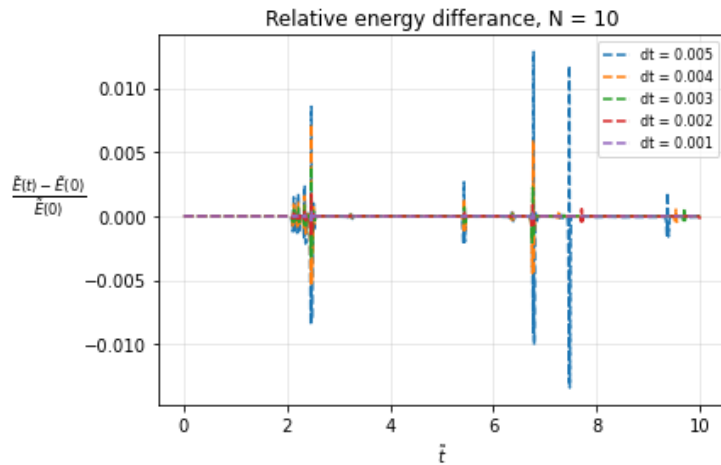


Figure 2: Relative energy difference for varying values of dt .

We will now increase the number of particles, and will also get more and more collisions. We therefore choose to set $dt = 0.001$ to try to minimize the spikes in energy, and increase the radius to $\tilde{R} = 15$ to ensure enough space between the particles in the starting position. The result of which can be found in figure 3.

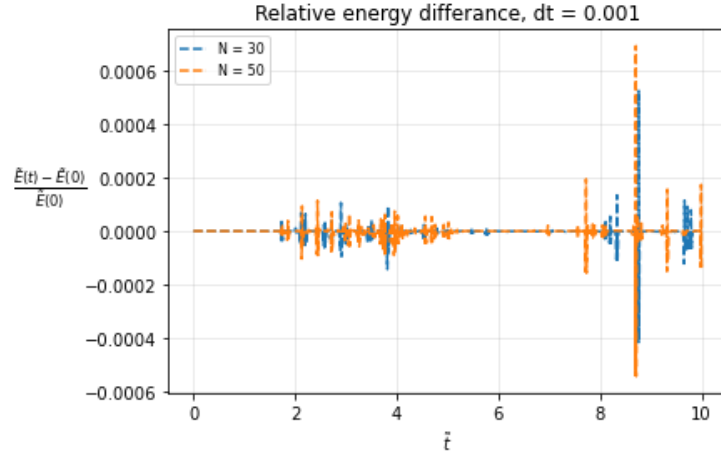


Figure 3: Relative energy difference for $dt = 0.001$ and varying N .

We will use the combination $dt = 0.001$ and $N = 30$ to further analyze the system.

3 Maxwell distribution & energy

To verify the Maxwell distribution, which states that the velocity statistics will follow

$$p(v_x) = \sqrt{\frac{m}{2\pi k_B T}} \exp\left(-\frac{mv_x^2}{2k_B T}\right), \quad (14)$$

given the particles behave as an ideal gas. For this to take place, we had to decrease the initial velocity of the particles from $v_{x0} = 5$ down to $v_{x0} = 2$.

To gather the statistics on the velocity we ran the simulation for a while, and sampled v_x of the particles at regular time-intervals after the system has settled. We then fitted a normal distribution to the data, and the result of this can be found in figure 4.

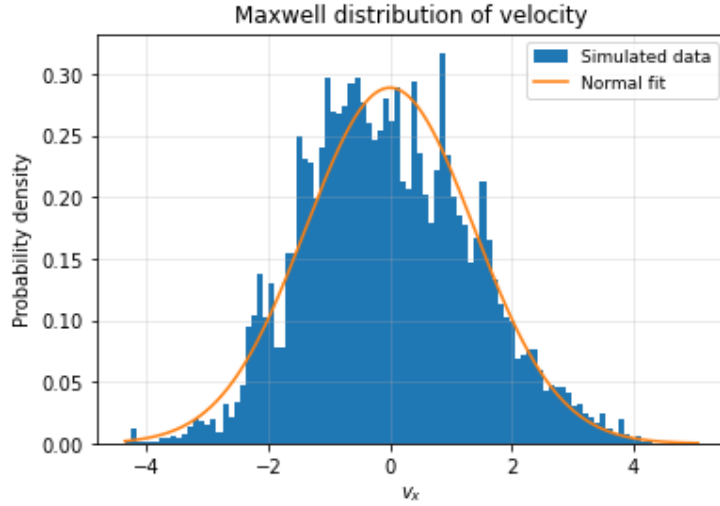


Figure 4: Velocity statistics of the simulated system of $N = 30$ particles.

In the code used to get the histogram, we also calculate the variance σ^2 for our distribution, and since it is centered at approximately zero and we have used dimensionless velocities, we get the relation

$$k_b T = \sigma^2. \quad (15)$$

The simulated result was $k_b T = 1.905547134325646$. This is in units of ϵ . Lastly, we calculated the energies throughout the simulation, and the result is displayed in figure 5. As we see, the potential is substantially smaller than the kinetic energy, and our assumption of ideal gas is justified.

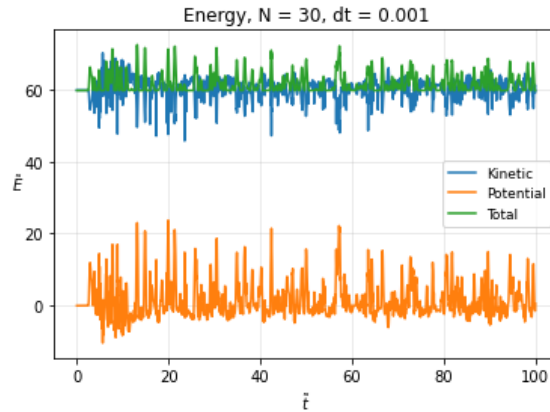


Figure 5: Kinetic, potential, and total energy of the simulated system of $N = 30$ particles.

4 Pressure & ideal gas

To calculate the time averaged pressure of the gas, we simply reused the particle-wall interaction part of the right-hand side of newtons equation, and divided the reaction force of the wall by the circumference of the circle. This, together with the result of our Maxwell distribution was used to calculate respectively the left and right-hand side of the ideal gas law for our system. We gradually lowered the energy, to see how the gas law holds up. The result of this analysis can be found in figure 6, and we clearly see that the absolute value of the relative error increase as the energy decreases.

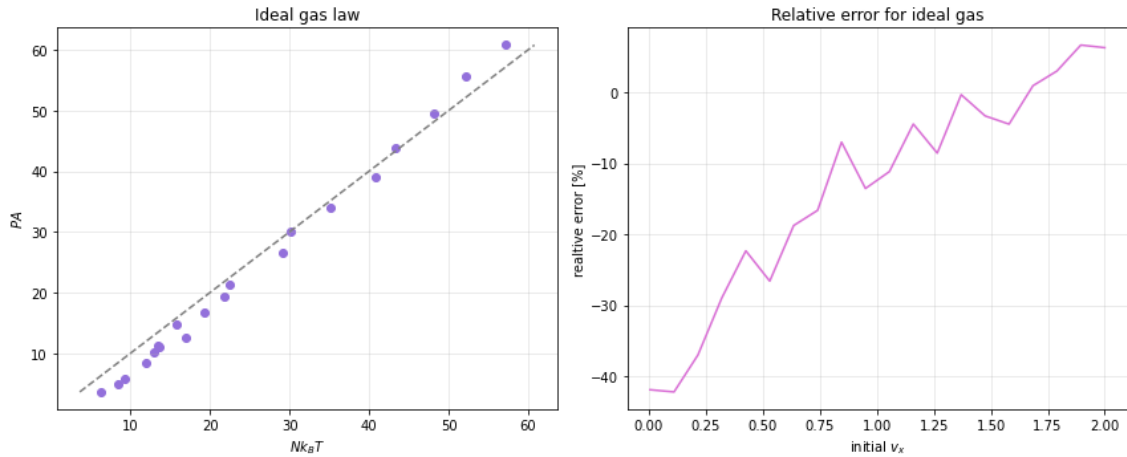


Figure 6: In the first figure we have a scatter plot of the value of PA against Nk_bT , with the diagonal $PA = Nk_bT$. The second shows the relative difference between PA and Nk_bT .

At small starting velocities, we see a big relative difference, of about 40% in absolute value. This indicates that the ideal gas law is no longer accurate, and we may suspect the gas has condensed into a mini droplet, a mentioned in the exercise sheet.